

# SIMATS ENGINEERING

## CO3 - ASSESSMENT TOOL 1

### Case Study

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

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#### COURSE OUTCOME COVERED

**CO3:** Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Assessment Tool Weightage: 25%

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#### QUESTIONS:

**Total Marks: 25**

#### Case Study 1: CNN for Automated Medical Image Classification

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

#### Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

## Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

### Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

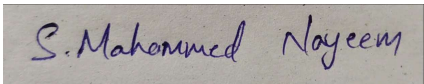
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### Code of Conduct:

*I S.Mahammed Nayeem /192425294 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**



**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>         | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                       | <b>Level 3 – Proficient</b>                    | <b>Level 2 – Developing</b>              | <b>Level 1 – Beginning</b>                           |
|-------------------------|------------------|------------------------------------------------------------------|------------------------------------------------|------------------------------------------|------------------------------------------------------|
| Understanding of Case   | 25               | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding ; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25               | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts          | Fails to apply relevant concepts                     |
| Analysis                | 20               | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth        | Weak or no analysis; lacks reasoning                 |
| Solution Design         | 20               | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification  | Incorrect or no solution provided                    |
| Clarity                 | 10               | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure            | Poor clarity; difficult to understand                |

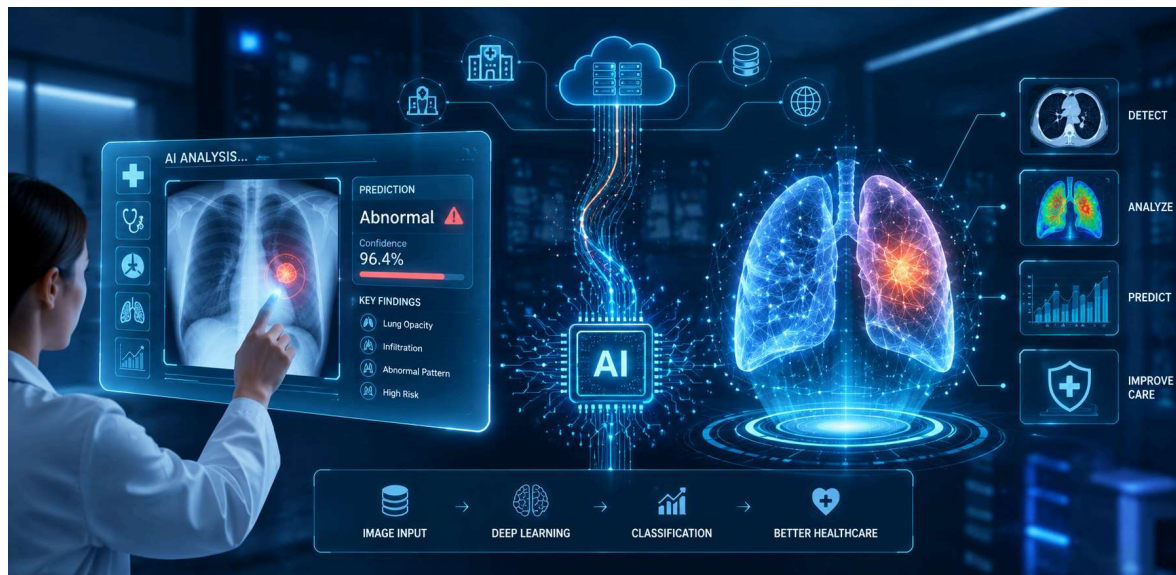
# Case Study 1: CNN for Automated Medical Image Classification

## 1. Introduction

The healthcare organization aims to classify chest X-ray images into **Normal** and **Disease Affected** categories. Since X-ray images may vary in size, brightness and quality, a **Convolutional Neural Network (CNN)** is suitable because it automatically learns important visual features such as edges, textures, shapes and abnormal regions.

## 2. Proposed CNN Architecture

A suitable architecture is:



### Main Layers

- **Input Layer:** Accepts resized and normalized X-ray images.
- **Convolutional Layers:** Extract important image features.
- **ReLU Activation:** Introduces non-linearity.
- **Pooling Layers:** Reduce image dimensions and computation.
- **Batch Normalization:** Stabilizes and improves training.
- **Global Average Pooling:** Converts feature maps into a compact representation.
- **Dropout:** Reduces overfitting.
- **Fully Connected Layer:** Performs final classification.
- **Output Layer:** Uses sigmoid to classify the image as Normal or

Disease-Affected. **Feature Learning**

Image → Edges → Textures → Shapes → Abnormal Patterns → Classification

## 3. Motivation for Using CNN

CNNs are preferred because they are specifically designed for image processing. **Advantages**

- Automatically extracts features from images.
- Preserves spatial relationships between pixels.
- Detects features at different locations.
- Requires less manual feature engineering.
- Learns hierarchical features from simple to complex patterns.
- Provides high classification performance for image-based tasks.

#### 4. Role of Filters, Convolution, Pooling and Fully Connected Layers

##### Convolution and Filters

A **filter/kernel** is a small matrix, usually  $3 \times 3$ , that slides across the image. It detects features such as:

- Edges
- Lines
- Textures
- Shapes
- Abnormal patterns

The output of convolution is called a **feature map**.

##### Pooling

Pooling reduces the size of feature maps while retaining important information. **Max pooling** selects the maximum value from a small region.

Benefits:

- Reduces computation
- Reduces memory usage
- Helps control overfitting
- Provides some translation tolerance

##### Fully Connected Layer

The fully connected layer uses the extracted features to make the final classification. **Output:**

Normal  $\rightarrow 0$

Disease-Affected  $\rightarrow 1$

#### 5. Parameter Sharing

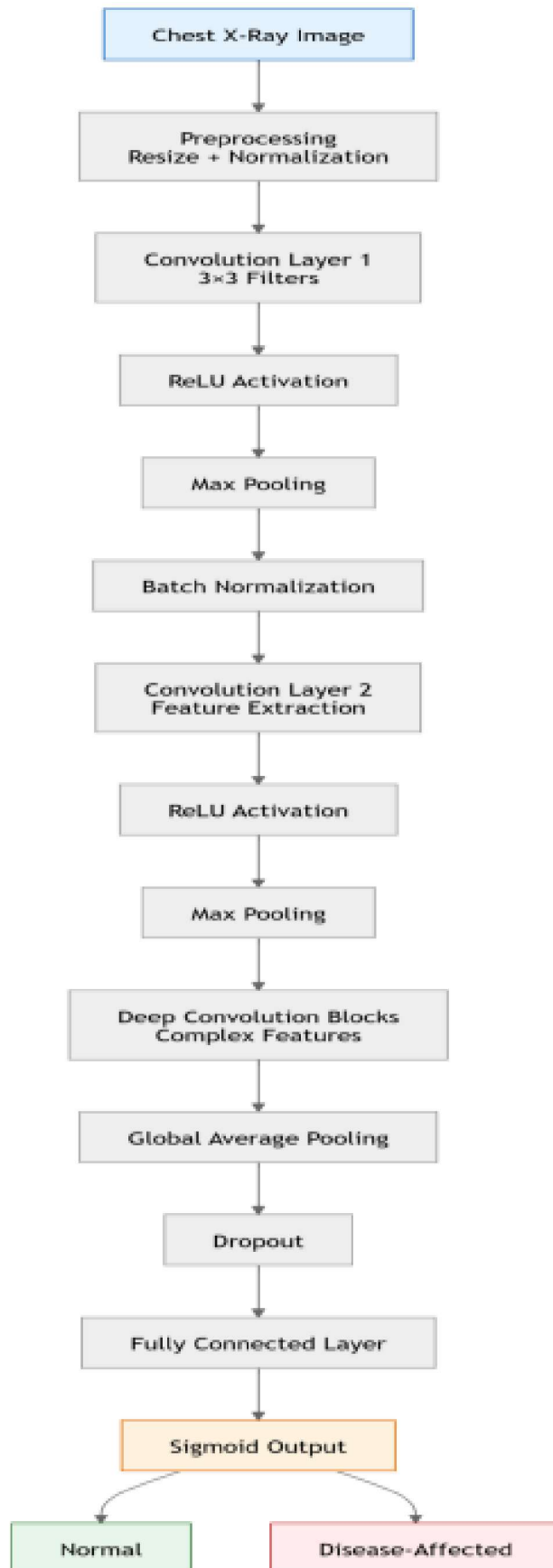
CNNs use the **same filter weights at different locations** of an image. This is called parameter sharing.

For example, a  $3 \times 3$  filter requires only **9 weights + 1 bias**, and the same filter

can scan the complete image.

In contrast, a fully connected network requires separate weights for individual pixel connections. **Benefits**

- Greatly reduces the number of parameters.
- Reduces memory and computational requirements.
- Helps detect the same feature at different locations.
- Improves generalization.



## 6. Regularization Techniques

Medical image datasets can cause overfitting if the model memorizes training images instead of learning general patterns.

### Data Augmentation

Images can be modified using suitable:

- Rotation
- Scaling
- Cropping
- Brightness/contrast changes

This increases image diversity.

### Dropout

Dropout randomly disables some neurons during training, preventing the network from depending too much on specific neurons.

### Batch Normalization

Batch normalization normalizes intermediate activations and makes training more stable and faster.

### Other Techniques

- Early stopping
- L2 regularization
- Learning-rate scheduling

## 7. AlexNet vs ResNet

| Feature                | AlexNet            | ResNet           |
|------------------------|--------------------|------------------|
| Depth                  | Relatively shallow | Can be very deep |
| Feature learning       | Good               | Strong           |
| Skip connections       | No                 | Yes              |
| Vanishing gradient     | More susceptible   | Reduced          |
| Training deep networks | Difficult          | Easier           |
| Performance            | Good               | Generally better |
| Medical application    | Suitable           | More suitable    |

### Recommended Architecture: ResNet

**ResNet is preferred** because its residual/skip connections make deep networks easier to train and help overcome the vanishing-gradient problem.

It can learn complex features from X-ray images and generally provides stronger feature representations than AlexNet.



For practical implementation, a moderate model such as **ResNet-18 or ResNet-34** can provide a good balance between accuracy and computational requirements.

flowchart TD

### **Conclusion**

CNNs are highly suitable for automated chest X-ray classification because they automatically learn meaningful visual features. **Convolution and filters** perform feature extraction, **pooling** reduces computational complexity, **parameter sharing** reduces the number of trainable parameters, and **dropout, augmentation and batch normalization** help prevent overfitting. Among AlexNet and ResNet, **ResNet is the better choice** because of its deeper feature-learning

# Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

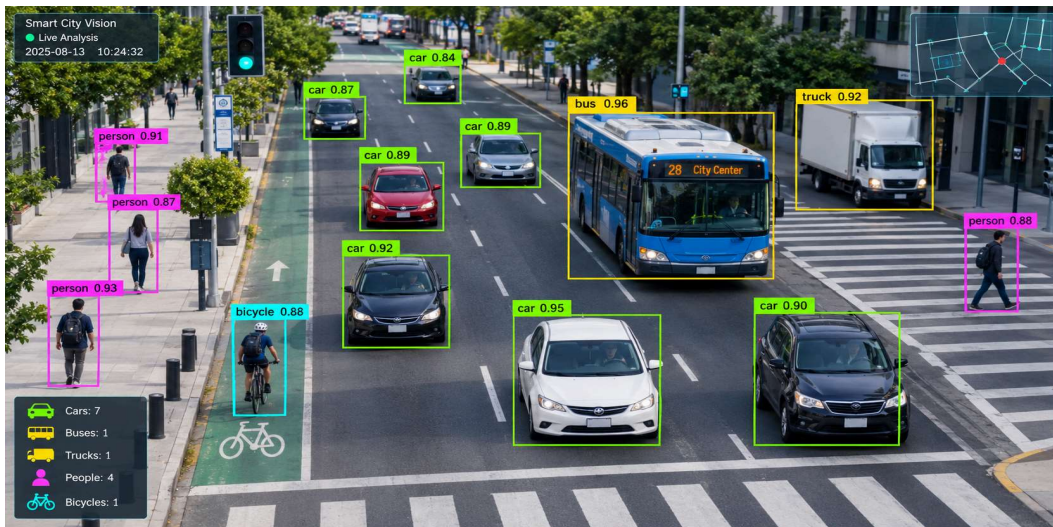
## 1. Introduction

An autonomous vehicle must identify objects such as **cars, pedestrians, bicycles, traffic signs and road obstacles** from camera images. Since the environment changes continuously due to lighting, distance, weather and viewing angle, a robust CNN-based vision system is required.

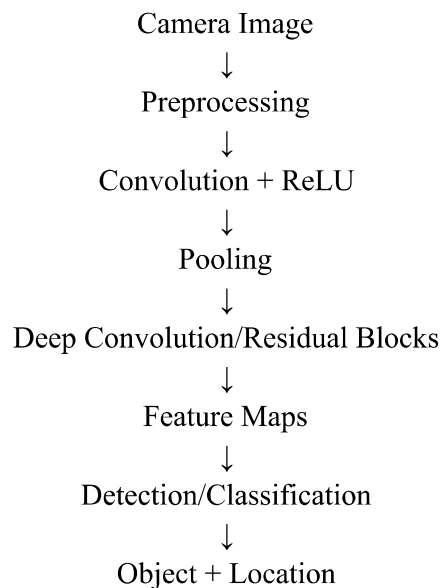
## 2. Proposed CNN Architecture

A suitable architecture is:

The system can be designed using a **ResNet backbone** for feature extraction.



## Basic Flow



For autonomous vehicles, **object detection** is more appropriate than simple classification because the system needs to identify **what the object is and where it is located**.

### 3. Role of CNN Layers

#### Input Layer

Receives camera images and performs preprocessing such as:

- Resizing
- Normalization
- Noise reduction

#### Convolutional Layer

Uses filters to detect visual patterns.

Early layers detect:

- Edges
- Lines
- Corners

Deeper layers detect:

- Wheels
- Vehicles
- Human shapes
- Traffic signs
- Road obstacles

#### ReLU Activation

Adds non-linearity and allows the network to learn complex patterns.

#### Pooling

Reduces spatial dimensions and computational requirements while retaining important features. **Output Layer**

Produces object classes and, in an object-detection system, their locations using bounding boxes.

### 4. Progressive Feature Learning

CNN filters learn features progressively.

Pixels

↓

Edges and Lines

↓

Textures and Shapes

↓

Object Parts

↓

Complete Objects

↓

Object Recognition

For example:

**Early layer → Edge of a car**

**Middle layer → Wheel/window**

**Deep layer → Complete car**

This hierarchical learning allows CNNs to recognize complex objects.

### **5. Parameter Sharing**

The same convolutional filter is applied across different locations of the image. For example, a filter that detects a vertical edge can detect that edge whether it occurs on the **left, center or right side** of the image.

#### **Benefits**

- Reduces trainable parameters.
- Reduces memory and computational cost.
- Allows features to be detected at different locations.
- Improves translation-related feature detection.
- Makes CNNs suitable for large camera images.

### **6. Overfitting and Regularization**

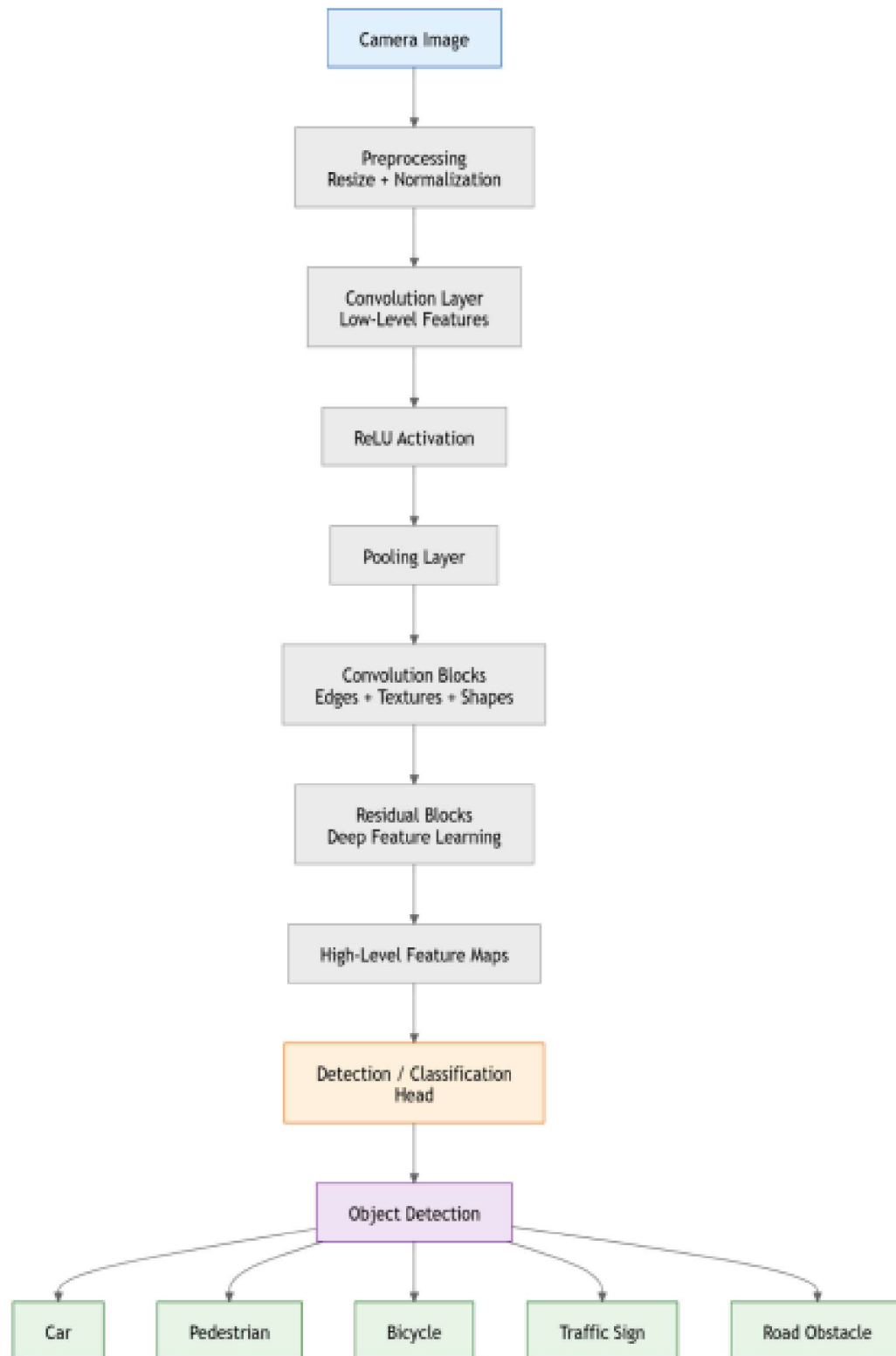
Overfitting can occur when the model performs well on training images but fails on new road conditions.

#### **Suitable techniques**

##### **Data Augmentation**

Use realistic variations such as:

- Brightness changes
- Scaling
- Cropping
- Rotation



### **Dropout**

Randomly removes neurons during training and reduces dependency on specific features.

### **Batch Normalization**

Improves training stability and convergence.

### Early Stopping

Stops training when validation performance stops improving.

### Diverse Dataset

Training data should include:

- Day and night
- Rain and clear weather
- Urban and rural roads
- Different camera angles
- Different object distances

## 7. AlexNet vs ResNet

| Feature                | AlexNet          | ResNet                |
|------------------------|------------------|-----------------------|
| Architecture           | Conventional CNN | Residual CNN          |
| Depth                  | Shallow          | Deep                  |
| Skip connections       | No               | Yes                   |
| Vanishing gradient     | More likely      | Reduced               |
| Feature learning       | Good             | Excellent             |
| Training deep networks | More difficult   | Easier                |
| Accuracy               | Good             | Generally higher      |
| Real-time use          | Lower complexity | Depends on model size |

## 8. Recommended Architecture

**ResNet is recommended for the autonomous vehicle system.**

### Reasons

#### High Accuracy:

ResNet can learn complex features required to distinguish pedestrians, vehicles, signs and obstacles.

#### Better Feature Learning:

Deep residual networks learn high-level semantic features effectively.

#### Reduced Vanishing Gradient:

Skip connections allow information and gradients to flow through deeper layers. **Real-Time Consideration:**

A moderate model such as **ResNet-18** or **ResNet-34** can provide a better balance between accuracy and computational cost than an unnecessarily deep model.

#### Detection Capability:

ResNet can be used as a feature-extraction backbone for an object-detection system.

### **Conclusion**

CNNs provide an effective solution for autonomous vehicle vision because they automatically learn hierarchical features from camera images. **Convolution and filters** identify visual patterns, **pooling** reduces computation, and **parameter sharing** enables efficient feature detection across different image locations.

Although AlexNet is computationally simpler, **ResNet is generally more suitable** because it provides stronger feature learning and reduces vanishing-gradient problems. A moderate ResNet model can achieve a practical balance between **accuracy, computational cost and real-time performance**.